

ALAMOGORDO WHITE SANDS, NM, USA

WMO: 722693

Lat: 32.840N

Lon: 105.991W

Elev: 4200

StdP: 12.60

Time Zone: -7.00 (NAM)

Period: 96-19

WBAN: 93097

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 20.9	(c) 24.9	(d) -1.9	(e) 5.8	(f) 42.4	(g) 0.7	(h) 6.6	(i) 45.2	(j) 24.8	(k) 52.5	(l) 21.0	(m) 53.3	(n) 3.5	(o) 140	(p) 0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 25.7	(c) 99.8	(d) 63.1	(e) 98.6	(f) 63.4	(g) 95.2	(h) 62.9	(i) 70.3	(j) 86.0	(k) 68.8	(l) 84.5	(m) 67.7	(n) 83.4	(o) 8.2	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	112.6	74.2	64.3	105.6	73.8	63.4	102.4	73.7	37.1	84.2	35.7	84.1	34.7	83.7	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.8	(b) 19.0	(c) 16.7	(e) 13.7	(f) 103.6	(g) 7.1	(h) 2.2	(i) 8.5	(j) 105.2	(k) 4.4	(l) 106.5	(m) 0.4	(n) 107.7	(o) -4.8	(p) 109.3		
(a) 22.8	(b) 19.0	(c) 16.7	(e) 11.9	(f) 71.3	(g) 6.1	(h) 1.9	(i) 7.5	(j) 72.6	(k) 3.9	(l) 73.7	(m) 0.5	(n) 74.7	(o) -3.9	(p) 76.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.4	43.4	47.7	54.7	61.8	70.1	79.6	80.7	79.3	74.0	62.4	50.7	43.0
	DBStd	15.10	6.63	7.62	7.20	6.63	6.16	4.63	3.86	3.83	4.83	7.36	7.34	7.13
	HDD50	710	224	118	39	4	0	0	0	0	0	7	81	237
	HDD65	2885	670	485	321	133	25	0	0	0	4	134	431	682
	CDD50	5221	18	53	186	358	623	888	953	909	720	392	101	20
	CDD65	1923	0	0	4	38	183	438	488	444	273	54	1	0
	CDH74	25429	0	12	192	828	2838	6266	6097	5270	3085	788	53	0
	CDH80	12896	0	0	23	198	1294	3637	3340	2776	1410	216	2	0
(14) Wind	WSAvg	6.7	5.8	6.9	7.6	8.5	8.0	7.6	6.6	6.0	5.9	6.2	5.7	5.9
(15) Precipitation	PrecAvg	12.6	0.7	0.6	0.4	0.2	0.5	0.9	2.2	2.4	1.9	1.3	0.7	0.8
	PrecMax	18.3	1.4	1.2	2.2	1.1	2.4	2.2	6.4	5.7	6.2	6.1	3.4	3.0
	PrecMin	5.9	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.3	0.0	0.0	0.0	0.0
	PrecStd	3.2	0.4	0.3	0.5	0.3	0.5	0.7	1.3	1.3	1.3	1.7	0.9	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.0	75.2	82.3	88.4	98.7	102.5	102.1	100.0	96.8	90.1	79.3	70.2
		MCWB	48.2	49.3	52.3	55.2	61.1	62.8	63.5	64.8	64.3	61.6	54.8	50.2
	2%	DB	65.6	72.1	79.2	84.2	93.1	100.1	99.4	97.5	93.1	84.5	74.9	65.7
		MCWB	46.2	48.1	50.6	53.9	58.4	61.8	64.4	65.7	63.1	59.0	52.6	48.3
	5%	DB	63.0	67.9	75.2	81.9	90.5	98.9	97.1	95.1	90.8	81.9	71.8	62.8
		MCWB	45.2	46.8	49.4	52.9	57.6	61.5	65.0	65.0	62.5	57.8	51.4	45.7
	10%	DB	58.6	63.8	72.6	79.4	87.8	95.2	93.5	92.7	88.4	78.9	66.2	57.3
		MCWB	42.8	45.2	48.5	51.9	56.6	60.4	65.0	65.1	62.3	56.7	48.9	43.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.7	52.9	56.2	60.5	65.3	69.7	73.3	71.8	69.6	66.9	58.4	52.4
		MCDB	62.7	65.0	73.3	79.2	87.0	90.7	91.7	87.3	83.7	78.3	72.2	65.4
	2%	WB	48.5	50.4	53.3	57.1	62.4	67.4	70.7	70.1	67.8	64.0	55.1	49.3
		MCDB	60.6	65.1	73.6	77.4	85.0	87.4	86.3	85.7	82.2	76.3	68.4	62.1
	5%	WB	46.4	48.6	51.2	55.0	60.3	65.7	68.9	68.7	66.4	61.4	52.9	46.8
		MCDB	59.2	63.5	71.5	76.2	83.4	87.0	84.0	84.4	81.0	74.7	66.4	58.9
	10%	WB	44.3	46.6	49.4	53.1	58.5	64.1	67.6	67.7	65.1	59.1	50.6	44.4
		MCDB	56.7	61.9	69.5	75.3	81.4	86.7	83.2	83.3	80.0	73.6	64.9	56.1
(35) Mean Daily Temperature Range	5% DB	MDBR	27.0	27.4	29.2	29.8	30.5	29.9	25.7	25.6	26.2	27.7	28.0	25.6
		MCDBR	31.1	32.5	33.2	32.6	33.8	32.7	29.3	29.0	29.8	30.9	32.7	29.8
		MCWBR	17.5	17.2	16.1	15.5	15.1	12.9	9.1	8.8	10.8	13.9	16.6	16.6
	5% WB	MCDBR	27.1	28.0	30.7	30.3	31.1	28.7	24.6	24.6	23.9	24.7	27.6	26.3
		MCWBR	16.0	15.4	15.5	15.3	14.0	11.3	8.2	8.4	9.0	11.7	15.1	15.2
(40) Clear-Sky Solar Irradiance	taub		0.251	0.259	0.279	0.307	0.324	0.348	0.389	0.372	0.334	0.291	0.270	0.256
	taud		2.606	2.579	2.512	2.417	2.424	2.390	2.349	2.368	2.438	2.552	2.561	2.593
	Ebn at Noon		315	321	319	311	304	295	283	287	295	304	304	306
	Edh at Noon		24	28	32	37	37	38	40	38	34	28	25	23
(44) All-Sky Solar Radiation	RadAvg		1126	1440	1860	2279	2479	2467	2198	2039	1838	1526	1202	1007
	RadStd		80	115	101	103	102	129	118	84	91	124	81	57

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+184	N/A

Nomenclature: See separate page